Dr. Priya Venkataraman

Associate Professor of Computational Biology

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Education

2008 - 2013 | **MIT**, Cambridge, MA
Ph.D. in Computational Biology, *Advisors:* Prof. Eric Lander and Prof. Manolis Kellis
`("Prof. Eric Lander", "Prof. Manolis Kellis")`

2004 - 2008 | **Indian Institute of Technology Bombay**, Mumbai, India
B.Tech. in Bioinformatics

Academic Positions

2020 - Present | **Harvard Medical School**, Boston, MA
Associate Professor

2020 - Present | **Broad Institute of MIT and Harvard**, Cambridge, MA
Associate Member

2015 - 2020 | **Stanford University**, Stanford, CA
Assistant Professor, Biomedical Data Science

2013 - 2015 | **UC Berkeley**, Berkeley, CA
Postdoctoral Fellow, *Supervisor:* Prof. Yun S. Song

Selected Publications

1. Venkataraman P., Chen R., Okafor J., & Srivastava K. (2024). Single-cell multi-omics reveals chromatin remodelling dynamics during T-cell exhaustion. *Nature Methods*, **21**(3), 412–428. doi:10.1038/s41592-024-02178-x
2. Okafor J., Venkataraman P., & Liang H. (2023). DeepSplice: a graph neural network for predicting tissue-specific alternative splicing. *Genome Research*, **33**(9), 1541–1556. doi:10.1101/gr.277901.123
3. Venkataraman P. & Kellis M. (2022). Regulatory grammar of enhancer–promoter loops across 200 human cell types. *Cell*, **188**(4), 987–1004. doi:10.1016/j.cell.2022.09.031
4. Chen R., Park S., & Venkataraman P. (2021). Epigenomic imputation with variational autoencoders across missing tissue types. *Nature Biotechnology*, **39**(7), 838–851. doi:10.1038/s41587-021-00900-y
5. Venkataraman P., Song Y.S., & Durbin R. (2019). Bayesian nonparametric clustering of structural variants in long-read sequencing. *PLOS Genetics*, **15**(6), e1008271. doi:10.1371/journal.pgen.1008271

Grants & Funding

2023 - 2028 | **Decoding Cis-Regulatory Logic of Cell-Fate Transitions**
NIH R01 HG012345, Principal Investigator

2022 - 2027 | **NIH Director's New Innovator Award DP2 HG011901**
National Institutes of Health, Principal Investigator

2021 - 2024 | **Scalable Deep Learning for Single-Cell Multi-Omics Integration**
NSF DBI-2145012, Principal Investigator

Awards & Honors

2022 | **NIH Director's New Innovator Award**
National Institutes of Health

2021 | **ISCB Overton Prize**
International Society for Computational Biology

2018 | **Sloan Research Fellowship in Computational & Evolutionary Molecular Biology**
Alfred P. Sloan Foundation

2017 | **NSF CAREER Award**
National Science Foundation

2013 | **Outstanding Doctoral Thesis Award**
MIT Department of Biology

Teaching

2021 - Present | **BCMP 363: Computational Genomics (graduate)**
Harvard Medical School

2020 - Present | **BCMP 200: Foundations of Data Science in Biology**
Harvard Medical School

2016 - 2019 | **BIODS 253: Algorithms for Genomics (graduate)**
Stanford University

Professional Service

- **Associate Editor** — *PLOS Computational Biology*, 2022–Present
- **Program Committee** — RECOMB 2021, 2022, 2023, 2024; ISMB 2020–2024
- **Reviewer** — *Nature Methods*, *Genome Research*, *Bioinformatics*, *Cell Systems*
- **Faculty Search Committee** — HMS Department of Biomedical Informatics, 2023
- **Organizing Committee** — RECOMB-Seq 2022 (Workshop Chair)

Research Interests

- Computational epigenomics and chromatin biology
- Single-cell and spatial multi-omics integration
- Machine learning for regulatory genomics
- Structural variant detection in long-read sequencing
- Causal inference in gene regulatory networks

Skills

Computational: Python (PyTorch, JAX, scikit-learn), R (Bioconductor), C++, Snakemake, Nextflow

Genomics: ATAC-seq, ChIP-seq, Hi-C, 10x Genomics Multiome, PacBio/ONT long-read analysis

Statistics: Bayesian inference, variational autoencoders, graph neural networks, causal discovery

References

- Prof. Eric Lander — Ph.D. Advisor — MIT / Broad Institute
- Prof. Manolis Kellis — Ph.D. Co-Advisor — MIT CSAIL
- Prof. Yun S. Song — Postdoctoral Mentor — UC Berkeley